

Polynomial Regression

Machine Learning Assignment 1 | BT2024146 | 8 October 2026

This report describes two separate polynomial-regression problems using the assigned data. The official assignment sets maximum total degrees of 10 for var1 and 20 for var2, and evaluates predictions against hidden test labels using mean squared error (MSE) and R-squared. The test labels were not available for this work.

Problems and data

Problem	Input features	Target	Train / test rows	Max degree
var1	6: x1 to x6	Net Power Score (y)	1,000 / 1,000	10
var2	3: x1 to x3	Thermal Anomaly Score (y)	1,000 / 1,000	20

Feature descriptions from the assignment

var1: power-plant controls	var2: survey coordinates
x1 high-pressure steam valve; x2 condenser coolant flow; x3 reinjection pump pressure; x4 turbine blade pitch; x5 non-condensable gas exhaust valve; x6 steam inlet pressure.	x1 east-west offset; x2 north-south offset; x3 vertical depth offset from the basecamp.

Observed data checks

Problem	Feature range	Target range (mean; SD)	Missing	Duplicate rows: train / test
var1	[-1, 1]	-9.59 to 13.95 (0.76; 3.27)	0	0 / 2
var2	[-1, 1]	-29.85 to 39.75 (2.24; 6.80)	0	0 / 6

All supplied training and test feature columns match within each problem; no missing or non-finite values were found. Test feature duplicates (2 in var1 and 6 in var2) were retained in their original positions. The sample submission has 1,000 rows and one column named y.

Validation and model search

Each problem has 1,000 labeled rows and no time or group identifier requiring a structured split. The comparison therefore uses RepeatedKFold with five shuffled folds, repeated twice, and random seed 2026. Each fold trains on 800 rows and validates on 200; results are averaged over 10 validation folds. Repeating the shuffled partition reduces dependence on a single split while keeping the degree search computationally manageable.

5 folds x 2 repeats	800 train / 200 validation per fold	Shuffle = yes; seed = 2026
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Polynomial models

PolynomialFeatures expands all monomials whose total degree is at most the candidate degree. The model search covers var1 degrees 1-10 and var2 degrees 1-20. With six inputs, degree 10 has 8,007 non-constant terms; with three inputs, degree 20 has 1,770. These sizes make unregularized high-degree fits unstable or underdetermined, so Ridge regularization was evaluated at every degree.

Candidate	Search	Purpose
Ridge polynomial regression	All allowed degrees; alpha in 0.01, 0.1, 1, 10, 100, 1,000, 10,000, 100,000	Controls coefficient size and high-degree variance.
Unregularized polynomial least squares	var1 degrees 1-5; var2 degrees 1-14	Baseline where the feature count is below 800 training rows.

Leakage control and selection

Within each validation fold, the polynomial map is deterministic and the StandardScaler is fitted only on that fold's 800 training rows, then applied to its 200 validation rows. The validation code solves the same Ridge objective in dual form to reuse a fold's matrix factorization across alpha values; the final selected model is refit with a scikit-learn pipeline on all 1,000 labeled rows. MSE and R-squared are averaged over folds. The one-standard-error rule takes the lowest degree with mean validation MSE no more than one fold-to-fold standard error above the minimum; within that degree it takes the lowest-MSE candidate.

Model-selection results

Problem	Selected model	Terms	CV MSE	CV R-squared	Min CV MSE
var1	Ridge, degree 5, alpha 20	461	0.515157	0.951535	0.515157
var2	Ridge, degree 10, alpha 1	285	0.262251	0.994186	0.260706

Best Ridge alpha and validation scores at each degree; var1 is on the left and var2 on the right. MSE is mean squared error; R2 is R-squared. Green cells mark the selected degrees.

d1	alpha	MSE	R2	d2	alpha	MSE	R2
1	50	9.3998	0.1175	1	1	33.4131	0.2724
2	5	2.8267	0.7335	2	10	21.0831	0.5391
3	5	1.0170	0.9043	3	1	11.0045	0.7583
4	15	0.7419	0.9303	4	0.01	3.3031	0.9272
5	20	0.5152	0.9515	5	0.1	1.4935	0.9672
6	30	0.5652	0.9469	6	0.01	0.5265	0.9883
7	50	0.6733	0.9367	7	0.01	0.3597	0.9920
8	50	0.7650	0.9283	8	0.1	0.2801	0.9938
9	100	0.9143	0.9142	9	0.1	0.2730	0.9939
10	100	1.0199	0.9045	10	1	0.2623	0.9942
				11	1	0.2607	0.9942
				12	1	0.2619	0.9942
				13	1	0.2658	0.9941
				14	1	0.2745	0.9939
				15	1	0.2821	0.9938
				16	10	0.2862	0.9937
				17	10	0.2898	0.9936
				18	10	0.2953	0.9935
				19	10	0.2993	0.9934
				20	10	0.3067	0.9932

Problem	Best OLS degree	Validation MSE	Validation R-squared
var1	4	0.815519	0.923266
var2	8	0.284971	0.993656

Ridge had lower validation MSE than the best tested OLS candidate in both problems. The selected Ridge settings are recorded with fold scores and training scores in outputs/model_selection_var1.csv and outputs/model_selection_var2.csv.

Selection rationale and final predictions

Why these models were selected

var1: Degree 5 with Ridge alpha 20 achieved the lowest mean validation MSE (0.515157; R-squared 0.951535) among all tested settings. Its 461 monomials keep the model below the 800-row fold size. At degree 10, the best Ridge model's training MSE was lower (0.132 versus 0.200 at degree 5), while validation MSE was higher (1.020 versus 0.515), consistent with higher variance at the maximum degree.

var2: Degree 11 with alpha 1 had the minimum raw cross-validation MSE (0.260706; R-squared 0.994227). Degree 11 achieved the lowest raw cross-validation MSE. However, degree 10 was selected using the one-standard-error rule because its validation error was within one standard error of the minimum while requiring fewer polynomial terms. Degree 10 with alpha 1 scored 0.262251, below the one-standard-error cutoff of 0.268001, and uses 285 terms rather than degree 11's 363 terms. From degree 11 to 15, the best-per-degree training MSE fell from 0.169 to 0.152, while validation MSE rose from 0.261 to 0.282; this supports the lower-complexity one-standard-error choice.

Problem	Mean training MSE	Mean validation MSE	Mean training R-squared	Mean validation R-squared
var1	0.2001	0.5152	0.9813	0.9515
var2	0.1760	0.2623	0.9962	0.9942

Focused Ridge alpha refinement

The local alpha grids were var1: 1, 3, 5, 7, 10, 15, 20, 30, 50 and var2: 0.1, 0.3, 0.5, 0.7, 1, 1.5, 2, 3, 5. They used the same 10 validation folds and fold-local scaling. A candidate was adopted only with at least 1% lower mean CV MSE and a paired mean fold improvement greater than its paired standard error. Var1 alpha 20 met both safeguards, so the complete var1 degree search was rerun with the focused values added; var2 retained alpha 1.

Problem	Baseline alpha	Best local alpha	Local CV MSE	Paired improvement (SE)	Decision
var1	10	20	0.515157	0.023291 (0.0055)	Accepted
var2	1	0.7	0.261935	0.000316305 (0.000402)	Kept 1

Final prediction procedure

After selection, each pipeline was refit on all 1,000 labeled rows for its problem and applied to the corresponding 1,000 test feature rows without sorting. The written files were reopened and checked automatically: each has exactly one y column, 1,000 finite predictions, and no index column. The two outputs are outputs/BT2024146_pred_var1.csv and outputs/BT2024146_pred_var2.csv. No hidden-test score is reported.

Reproduction

From the project root, place the instructor-provided CSV files in data/, install requirements.txt, and run `python src/train_and_predict.py`. This recreates the CV tables, diagnostics, selection summary, and both prediction files. Run `python src/build_report.py` to regenerate this report. The project and training code are available in the provided GitHub repository: github.com/Varun576253/ML_Assignment.